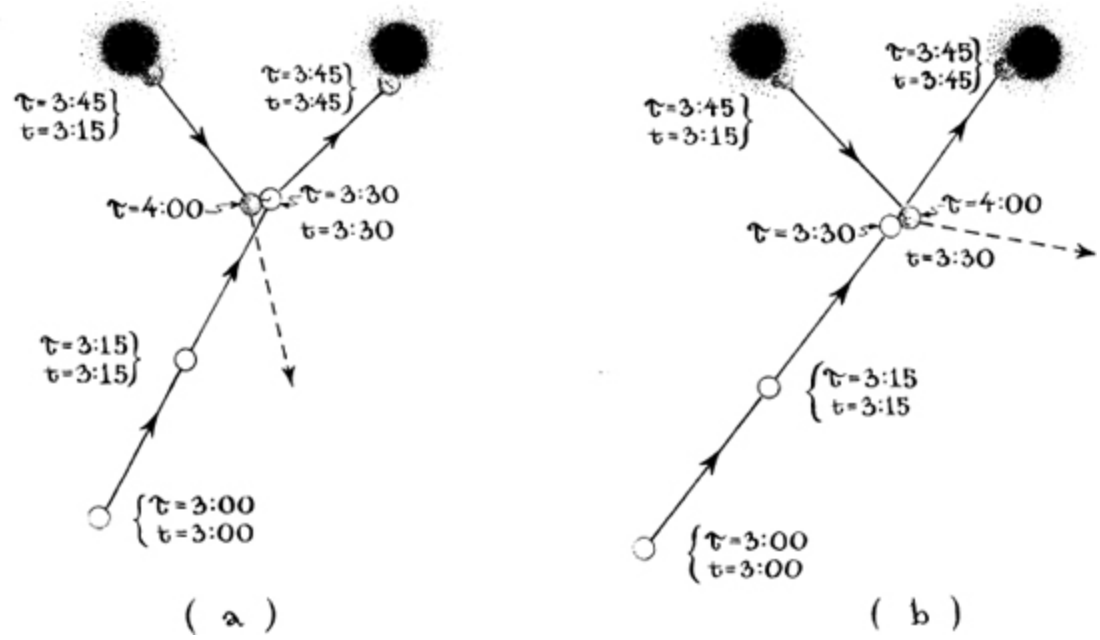
the ball's motion. The time machine has ruined this. There now are two, equally good predictions for the ball's motion.



14.10 The resolution of Polchinski's version of the matricide paradox (Figure 14.9): A billiard ball, starting out at 3:00 P.M. with the same initial conditions (same location and velocity) as in Polchinski's paradox, can move along either of the two trajectories shown here. Each of these trajectories is fully self-consistent and satisfies the classical laws of physics everywhere along the trajectory.

Actually, the situation is even worse than it looks at first sight: The time machine makes possible an *infinite number* of equally good predictions for the ball's motion, not just two. Box 14.2 shows a simple example.

Box 14.2

The Billiard Ball Crisis: An Infinity of Trajectories

One day, while sitting in San Francisco Airport waiting for a plane, it occurred to me that, if a billiard ball is fired between the two mouths of a wormhole-based time machine, there are two trajectories on which it can travel. On one (a), it hurtles between the mouths unscathed. On the other (b), as it is passing between the two mouths, it gets hit and knocked rightward, toward the right mouth; it then goes down the wormhole, emerges from the left mouth before it went down, hits itself, and flies away.